

IgA Specificity in the Infant Gut Is Shaped by Diet and Increasingly Targets Persistent Species

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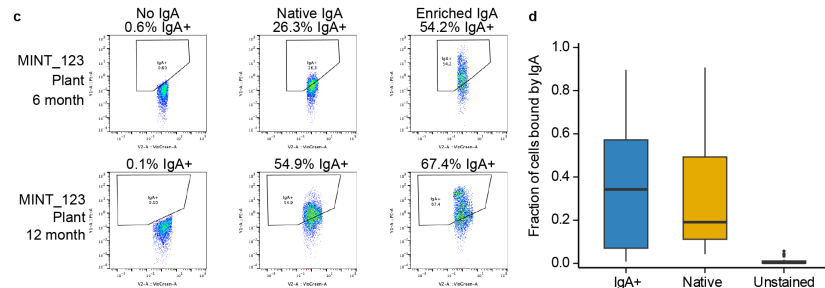
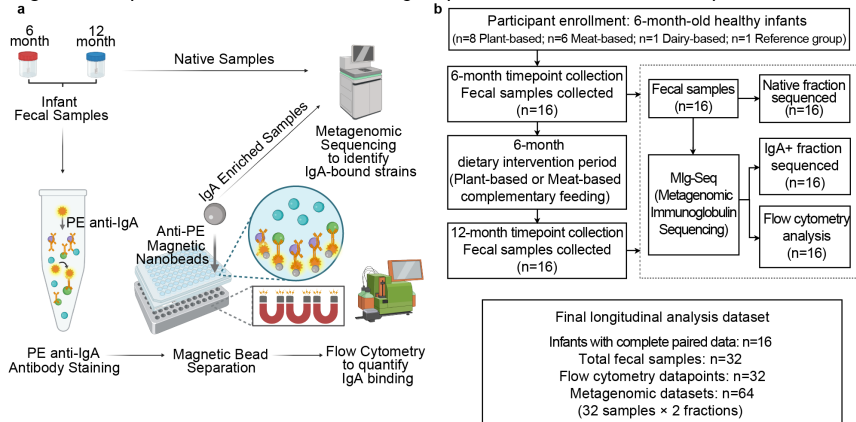
Introduction

- **The first year of life** is a critical window for immune and microbial development.
- Host-microbe symbiosis is mediated by **mucosal IgA**, the most abundant antibody in breast milk and the infant gut.
- **Diet** is a primary driver of the infant gut microbiome, particularly during complementary feeding.
- **Knowledge Gap:** The link between diet and immune targeting at the strain level is not well understood.

- **Our Study:** We optimized a new technique, Metagenomic Immunoglobulin Sequencing (**MIG-Seq**; *Nature Microbiology* 2025) for infant fecal samples and applied it to 32 samples from 16 infants in the **MINT trial**, a randomized controlled trial comparing **meat-, dairy-, plant-based, and reference complementary diets** from 5 to 12 months in both breast- and formula-fed infants. We tested whether different diets leave distinct strain-level IgA targeting "imprints" that may influence early-life host-microbe interactions.

Methods

Figure 1. Adaptation and validation of the MIG-Seq workflow for infant fecal samples.



Results

Figure 2. β -diversity analysis reveals age-driven community structure and enhanced individual specificity in the IgA-coated infant microbiome.

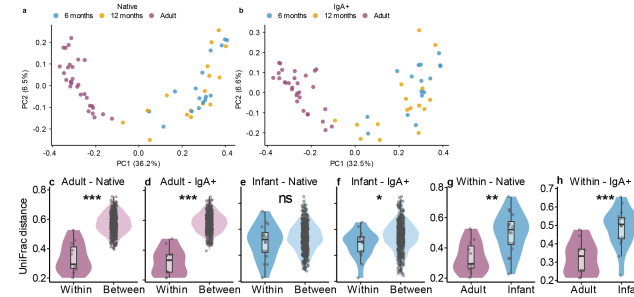


Figure 3. IgA targeting is conserved at the phylum level but diffuse at the species level in infants.

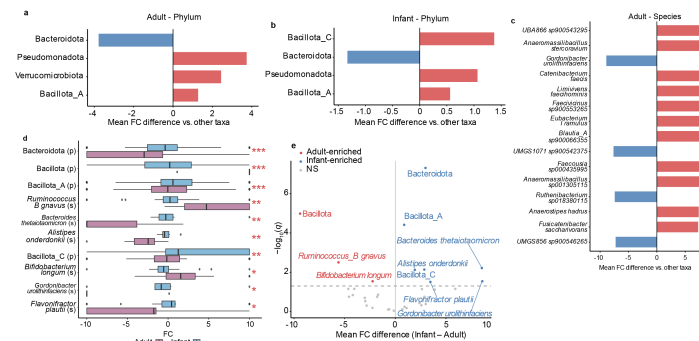


Fig 2 & 5. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; ns, not significant.
Fig 3 & 4. * $q < 0.1$, * $q < 0.05$, ** $q < 0.01$, *** $q < 0.001$; ns, not significant

Key Findings

- Successful MIG-Seq protocol for infants: Overcame low-biomass challenges in pediatric samples.
- Infant IgA targeting mirrors adult patterns at the phylum level but is more dynamic and less species-selective; it is more temporally stable than the microbiome itself.
- Early complementary diet shapes IgA targeting of specific taxa by 12 months, and this effect depends on concurrent breast milk exposure.
- IgA binding of persistent colonizers, especially *Bifidobacterium*, strengthens progressively during the 6–12 month window.

Figure 4. IgA targeting of specific taxa is influenced by feeding mode, dietary intervention, and their interaction.

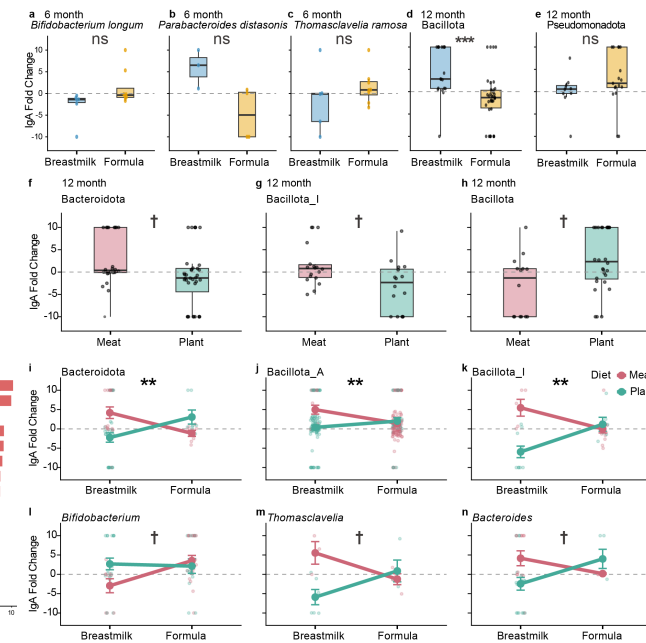
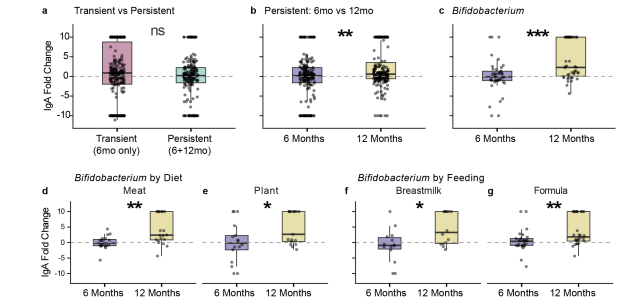


Figure 5. Persistent colonizers become progressively targeted by IgA.



Acknowledgements

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